

Copula-Based Anomaly Detection in Spatio-Temporal Seismic Data

1 Introduction

Seismic data, inherently spatio-temporal, require methods that account for dependencies among signals from multiple stations. Traditional univariate approaches for anomaly detection pool p -values but fail to consider spatial and temporal dependencies. Copulas, which decouple marginal distributions from dependency structures, offer a robust framework for addressing these challenges. This proposal outlines a methodology integrating spatio-temporal modeling, time series alignment, and copula-based pooling to improve anomaly detection accuracy.

2 Proposed Methodology

2.1 Spatio-Temporal VARIMA-X Model

To capture seismic dynamics, we model the observations $\mathbf{y}_t = [y_{1,t}, \dots, y_{k,t}]^\top$ from k stations using a VARIMA-X model:

$$(1 - B)^d \mathbf{y}_t = \sum_{i=1}^p \Phi_i (1 - B)^d \mathbf{y}_{t-i} + \sum_{j=1}^q \Theta_j \mathbf{w}_{t-j} + \Gamma \mathbf{x}_t + \mathbf{w}_t, \quad (1)$$

where B is the backshift operator, d the differencing order, Φ_i and Θ_j the autoregressive and moving average coefficients, $\mathbf{x}_t$ the exogenous covariates, and $\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ the error term.

2.2 Time Series Alignment with DTW

Dynamic Time Warping (DTW) aligns time series by minimizing cumulative distance under constraints:

$$\mathbf{W} = \arg \min_{\mathbf{W}} \sum_{(i,j) \in \mathbf{W}} |a_i - b_j|, \quad (2)$$

where $\mathbf{A}$ and $\mathbf{B}$ are the time series to be aligned. Alignment ensures residuals reflect true seismic anomalies.

2.3 Copula-Based p -Value Pooling

Using Sklar’s Theorem, the joint distribution of p -values $\mathbf{p}_t = (p_{1,t}, \dots, p_{k,t})$ is modeled via a copula C :

$$C(u_1, \dots, u_k) = F(F_1^{-1}(u_1), \dots, F_k^{-1}(u_k)), \quad (3)$$

where F is the joint distribution and F_i the marginals. The combined p -value is computed as:

$$p_{\text{combined},t} = 1 - C(p_{1,t}, \dots, p_{k,t}). \quad (4)$$

Anomalies are flagged if $p_{\text{combined},t} < \alpha$.

3 Theoretical Validation

3.1 Asymptotic Normality of Residuals

Under regular conditions, residuals from the VARIMA-X model converge to normality, ensuring valid hypothesis testing.

3.2 Copula-Based Dependency Modeling

Copulas effectively capture dependencies among p -values, enhancing sensitivity to anomalies while controlling false positives.

4 Simulation Study

Simulated data from a VARIMA-X model with induced anomalies demonstrate the proposed method’s superiority over traditional pooling techniques in terms of detection accuracy and false positive rates.

5 Case Study: Real Seismic Data

We will apply this framework to seismic data from multiple stations, highlighting its ability to detect anomalies corresponding to seismic events. DTW ensures alignment, while copulas model dependencies among station-specific p -values. The idea is to show with high confidence that pooling p -values from aligned seismic waves provides a better anomaly characterization than with one wave alone.

References